1. Criar a imagem:

```
FROM python:3.9-slim

WORKDIR /app

COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt

COPY . .

CMD ["python", "app.py"]
```

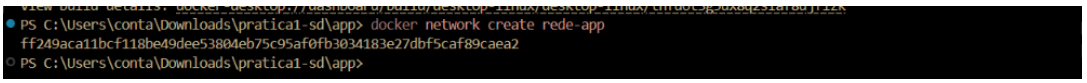
Dockerfile

2. Execução da aplicação utilizando a imagem escolhida:



Execução do comando docker build -t app-flask . no terminal

3. Executar um container Redis:

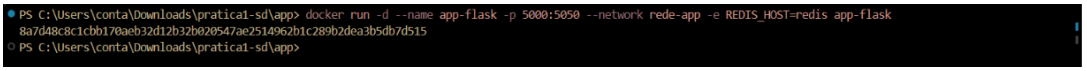


Criando a rede compartilhada com docker network create rede-app

4. Configurar comunicação entre os containers



Subindo o redis com o comando docker run -d –name redis –network rede-app redis:alpine



Subindo a aplicação flask, determinando a porta 5000:5050 e conexão com rede-app



Tela do navegador, com contador em 2 (carregado 2 vezes)



Limpando os containers atuais, para avançar para o docker-compose

5. Migrar para o docker-compose:

```yaml
version: '3.9'

services:
  redis:
    image: redis:alpine
    container_name: redis
    volumes:
      - redis-data:/data
    networks:
      - rede-distribuida
```

```yaml
  app:
    build: .
    container_name: app-flask
    ports:
      - "5000:5050"
    environment:
      - REDIS_HOST=redis
    depends_on:
      - redis
    networks:
      - rede-distribuida

volumes:
  redis-data:

networks:
  rede-distribuida:
```

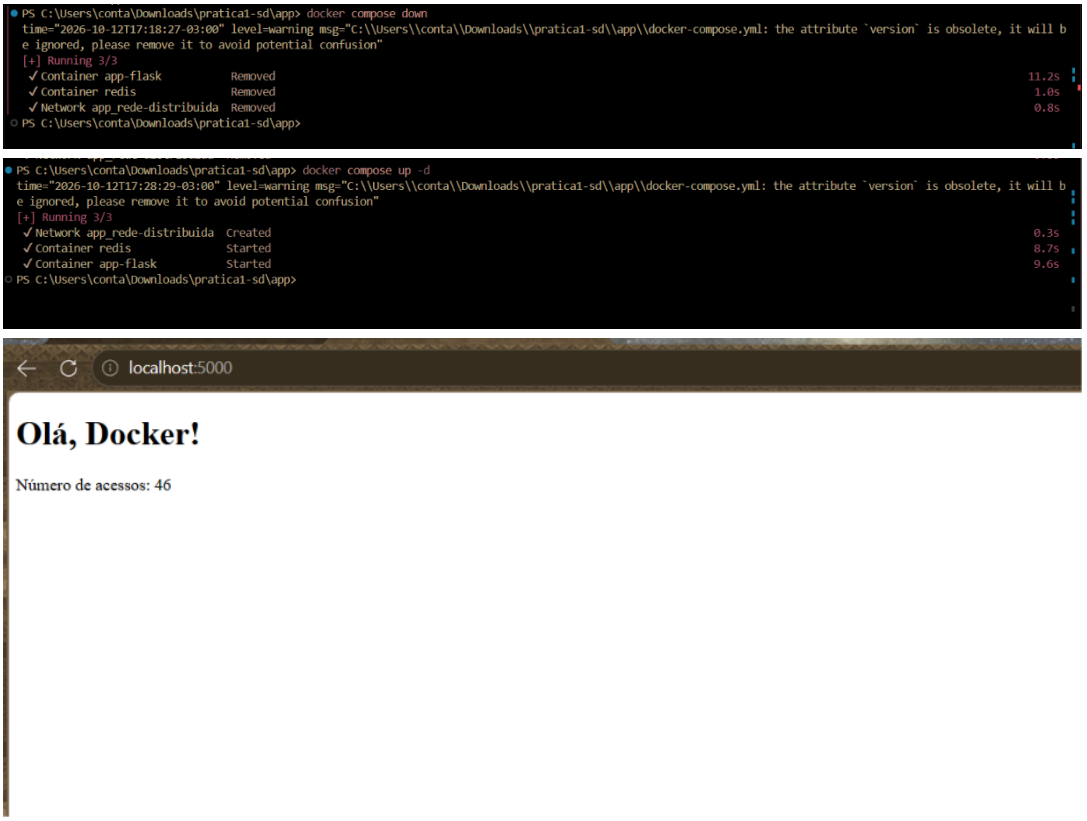
docker-compose.yml

6. Executar a aplicação com docker compose:

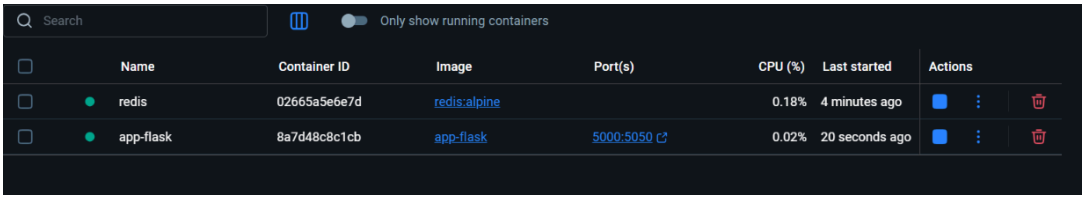


Testando compose com up, para ver a persistência.

7. Testar a persistência:



Testando compose com down+up, para ver a persistência



Containers em visualização na interface do docker

8. Modificando a aplicação:

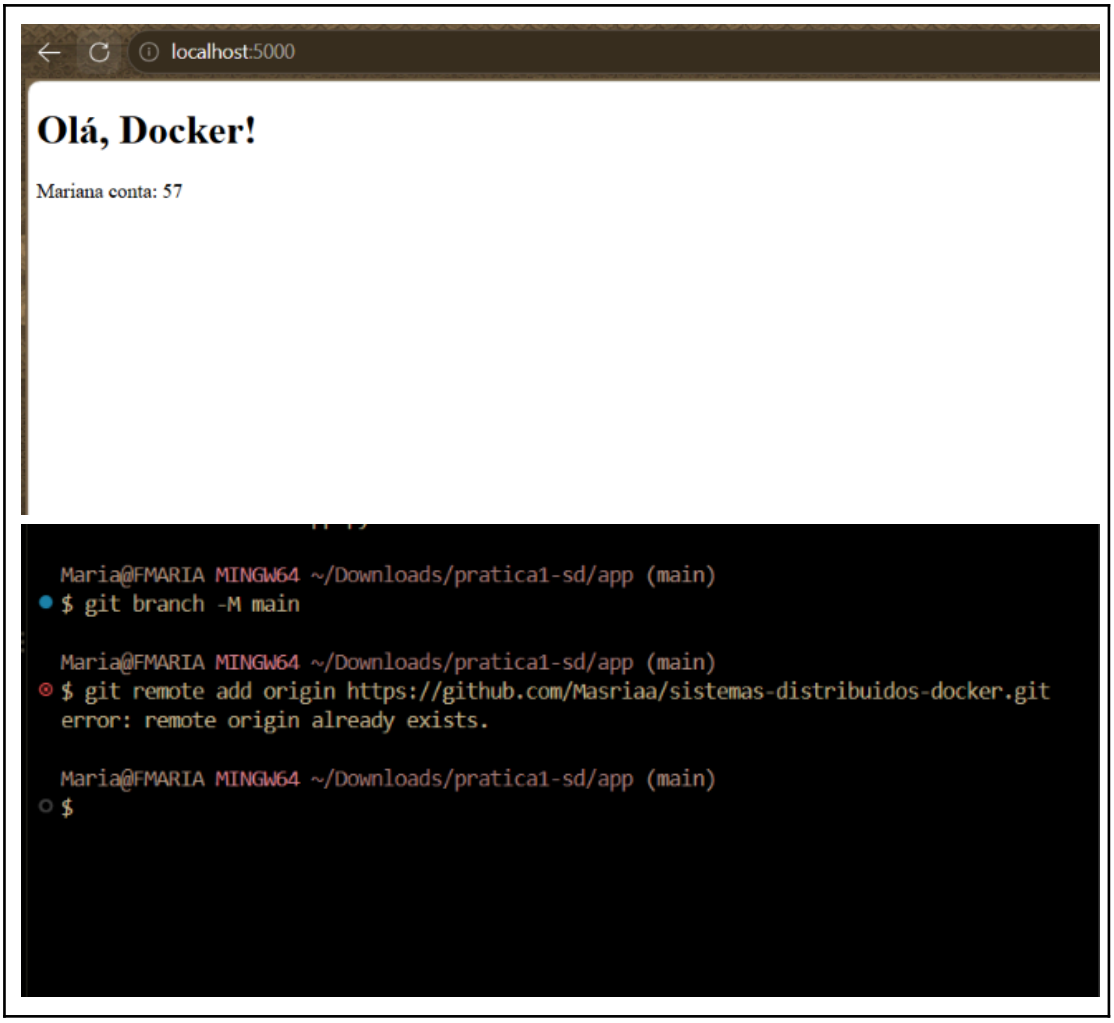
```
<!DOCTYPE html>
  <html lang="pt-BR">
  <head>
      <meta charset="UTF-8">
      <title>Olá, Docker!</title>
  </head>
  <body>
      <h1>Olá, Docker!</h1>
      <p>Mariana conta: {acessos}</p>
  </body>
```

```
    </html>
```

Trecho app.py modificado.

```
PS C:\Users\conta\Downloads\pratica1-sd\app> docker compose up -d --build
time="2026-10-12T17:43:55-03:00" level=warning msg="C:\\Users\\conta\\Downloads\\pratica1-sd\\app\\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion"
[+] Building 1.3s (12/12) FINISHED
 => [internal] load local bake definitions                                    0.0s
 => => reading from stdin 381B                                                0.0s
 => [internal] load build definition from Dockerfile                          0.0s
 => => transferring dockerfile: 194B                                          0.0s
 => [internal] load metadata for docker.io/library/python:3.9-slim            0.5s
 => [internal] load .dockerignore                                             0.0s
 => => transferring context: 2B                                               0.0s
 => [1/5] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acda1e13cf1731b1b  0.0s
 => [internal] load build context                                             0.0s
 => => transferring context: 4.18kB                                           0.0s
 => CACHED [2/5] WORKDIR /app                                                 0.0s
 => CACHED [3/5] COPY requirements.txt .                                      0.0s
 => CACHED [4/5] RUN pip install --no-cache-dir -r requirements.txt           0.0s
 => CACHED [5/5] COPY . .                                                     0.0s
 => exporting to image                                                        0.0s
 => => exporting layers                                                       0.0s
 => => writing image sha256:df84cf8161981aa32dd2b26bba7eb1d890d4695d33624e259fbd12492c8e2719  0.0s
 => => naming to docker.io/library/app-app                                    0.0s
 => resolving provenance for metadata file                                    0.0s
[+] Running 4/4
 ✓ app                          Built                                        0.0s
 ✓ Network app_rede-distribuida Created                                      0.1s
 ✓ Container redis              Started                                      0.7s
 ✓ Container app-flask          Started                                      0.9s
PS C:\Users\conta\Downloads\pratica1-sd\app>
```




localhost:5000
Olá, Docker!
Mariana conta: 57
Maria@FMARIA MINGW64 ~/Downloads/pratica1-sd/app (main)
$ git branch -M main
Maria@FMARIA MINGW64 ~/Downloads/pratica1-sd/app (main)
$ git remote add origin https://github.com/Masriaa/sistemas-distribuidos-docker.git
error: remote origin already exists.
Maria@FMARIA MINGW64 ~/Downloads/pratica1-sd/app (main)
$